

NATIONAL JUNIOR COLLEGE, SINGAPORE  
Senior High 2  
Preliminary Examination  
Higher 2

CANDIDATE  
NAME

BIOLOGY  
CLASS

REGISTRATION  
NUMBER

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## BIOLOGY

**9744/01**

Paper 1 Multiple Choice

11 September 2018

1 hour

Additional Materials: Multiple Choice Answer Sheet

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### READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Biology class and registration number above and on the Answer Sheet provided.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

**Read the instructions on the Answer Sheet very carefully.**

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

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This document consists of **14** printed pages.

**[Turn over**

- 1 Most wild plants contain toxins that deter animals from eating them. A scientist discovered that a toxin produced by a certain plant was also toxic to the same plant if it were applied to the roots of the plant. To find out why the plant was not normally killed by its own toxin, he fractionated some plant cells and found that the toxin was in the fraction that contained the largest cell organelle. He also found that the toxin was no longer toxic after it was heated.

Which statements are consistent with the scientist's observations?

- 1 The toxin is stored in the central vacuole.
- 2 The toxin cannot cross the membrane of the organelle in which it is stored.
- 3 The toxin is stored in chloroplast.
- 4 The toxin is likely to be lipid-soluble.
- 5 The toxin may be an enzyme.

- A** 1, 2 and 5  
**B** 1, 4 and 5  
**C** 2, 3 and 4  
**D** 3, 4 and 5

- 2 The distribution of membranes in three cell types – a prokaryotic cell, a liver cell and an enzyme-secreting cell from the stomach – is determined. For each cellular structure, the amount of membrane is expressed as a percentage of the total amount of membrane in the cell.

Which row of values best represents the membrane distribution in the three cell types?

CM: cell membrane

rERM: rough endoplasmic reticulum membrane

IMM: inner mitochondrial membrane

OMM: outer mitochondrial membrane

|          | prokaryotic cell |      |     | liver cell |     |     | stomach cell |     |     |
|----------|------------------|------|-----|------------|-----|-----|--------------|-----|-----|
|          | CM               | rERM | IMM | rERM       | OMM | IMM | rERM         | OMM | IMM |
| <b>A</b> | 99               | 0    | 0   | 35         | 7   | 28  | 60           | 16  | 4   |
| <b>B</b> | 99               | 0    | 0   | 35         | 7   | 28  | 60           | 4   | 16  |
| <b>C</b> | 52               | 12   | 36  | 60         | 7   | 28  | 35           | 4   | 16  |
| <b>D</b> | 52               | 12   | 36  | 60         | 28  | 7   | 35           | 4   | 16  |

- 3 A certain cell surface membrane is made entirely of phospholipids and is 7 nm thick. A volume of 1 mm<sup>3</sup> of this membrane was homogenised and dropped onto the surface of water in a large tray. The phospholipids spread out to form a continuous thin film.

What is the expected surface area of this film?

- A 143 000 mm<sup>2</sup> because the phospholipids formed a single layer  
 B 143 000 mm<sup>2</sup> because the phospholipids formed a double layer  
 C 286 000 mm<sup>2</sup> because the phospholipids formed a single layer  
 D 286 000 mm<sup>2</sup> because the phospholipids form a double layer
- 4 Magnesium ions are usually added to a polymerase chain reaction in the form of magnesium chloride. The reaction rate of *Taq* polymerase decreases as the concentration of magnesium chloride is reduced.

What is the role of the magnesium ions?

- A co-enzyme for *Taq* polymerase  
 B co-factor for *Taq* polymerase  
 C competitive inhibitor of *Taq* polymerase  
 D non-competitive inhibitor of *Taq* polymerase
- 5 A peptide consists of ten amino acids of four different kinds.

What is the number of tRNA molecules required to translate the mRNA for this peptide?

- A 4  
 B 10  
 C 12  
 D 30
- 6 Which row best describes the functions of the enzymes involved in DNA replication?

|   | unwinding of the DNA molecules | assembly of the leading strand | filling in of gaps between new DNA fragments | fusing together of new DNA fragments |
|---|--------------------------------|--------------------------------|----------------------------------------------|--------------------------------------|
| A | polymerase                     | ligase                         | polymerase                                   | helicase                             |
| B | helicase                       | polymerase                     | polymerase                                   | ligase                               |
| C | ligase                         | polymerase                     | helicase                                     | polymerase                           |
| D | helicase                       | polymerase                     | ligase                                       | polymerase                           |

- 7 The packing of DNA in the nucleus is necessary to compact the DNA to fit within the nucleus.

The following statements describe this process.

- 1 Looped domains are formed with the aid of chromosome scaffold.
- 2 Further coiling results in formation of condensed chromatin as seen in metaphase.
- 3 DNA winds twice around a histone octamer to form nucleosome.
- 4 Subsequent coiling results in the formation of a solenoid fibre.

Which combination correctly describes the sequence of DNA packing?

- A** 1, 3, 4, 2  
**B** 1, 4, 3, 2  
**C** 3, 1, 4, 2  
**D** 3, 4, 1, 2

- 8 Which row correctly describes the transfer of DNA from one bacterium to another?

|          | binary fission                                             | transduction                                                   | conjugation                                                          |
|----------|------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------------|
| <b>A</b> | bacterial chromosome and plasmids passed to daughter cells | DNA transferred by bacteriophage from one bacterium to another | single strand of F plasmid transferred from one bacterium to another |
| <b>B</b> | bacterial chromosome and plasmids passed to daughter cells | bacterium takes up foreign DNA from culture medium             | double-stranded F plasmid transferred from one bacterium to another  |
| <b>C</b> | only plasmids passed to daughter cells                     | DNA transferred by bacteriophage from one bacterium to another | single strand of F plasmid transferred from one bacterium to another |
| <b>D</b> | only plasmids passed to daughter cells                     | bacterium takes up foreign DNA from culture medium             | double-stranded F plasmid transferred from one bacterium to another  |

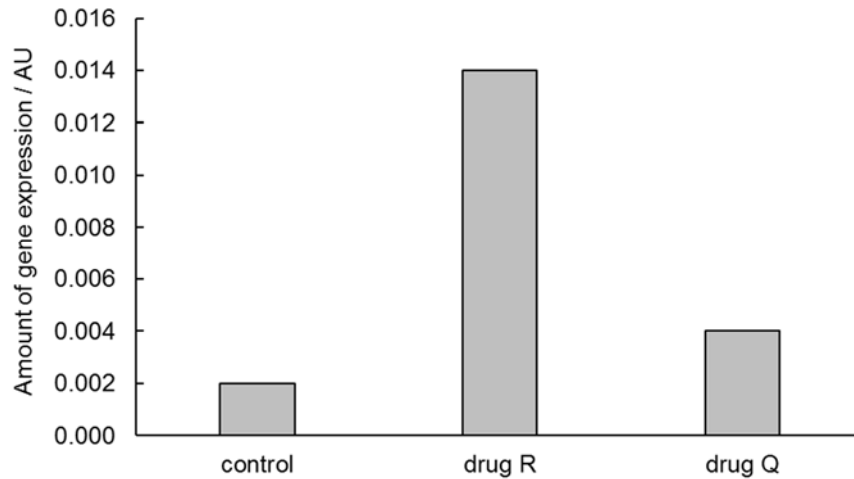
- 9 In an experiment, influenza A virus was engineered to have H1N1 envelope with H5N2 genome.

When the virus reproduces, what would be expected in the progeny?

- A** H1N1 envelope with H5N2 genome  
**B** H5N2 envelope with H1N1 genome  
**C** H5N1 envelope with H5N1 genome  
**D** H5N2 envelope with H5N2 genome

- 10** Drug R is a DNA methyltransferase inhibitor and drug Q is a histone deacetylase inhibitor. An experiment was carried out to investigate the effects of drug R and Q on the expression of a gene.

The graph shows the experimental results.



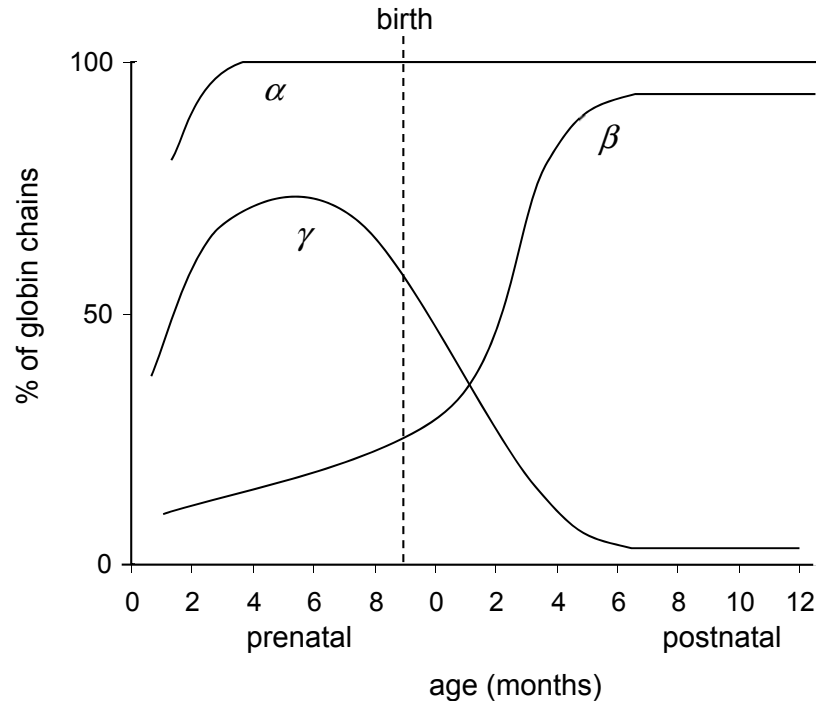
Which statements are possible explanations for the results shown?

- 1 Drug R increases gene expression by preventing methylation at CpG islands at the promoter.
- 2 Inhibiting DNA methylation is more effective in increasing gene expression than inhibiting histone deacetylation.
- 3 Drug Q results in weaker binding of histones to DNA.
- 4 Drug Q increases gene expression by increasing the accessibility of RNA polymerase to the promoter.

- A** 2 and 4 only  
**B** 3 and 4 only  
**C** 1, 3 and 4  
**D** 1, 2, 3 and 4

- 11** The globin gene family in humans consists of  $\alpha$ ,  $\beta$  and  $\gamma$  genes. These genes code for the globin chains that make up haemoglobin and are expressed at different levels during different developmental stages.

The graph shows the expression of the various globin chains during the prenatal (fetal) and postnatal (after birth) periods.

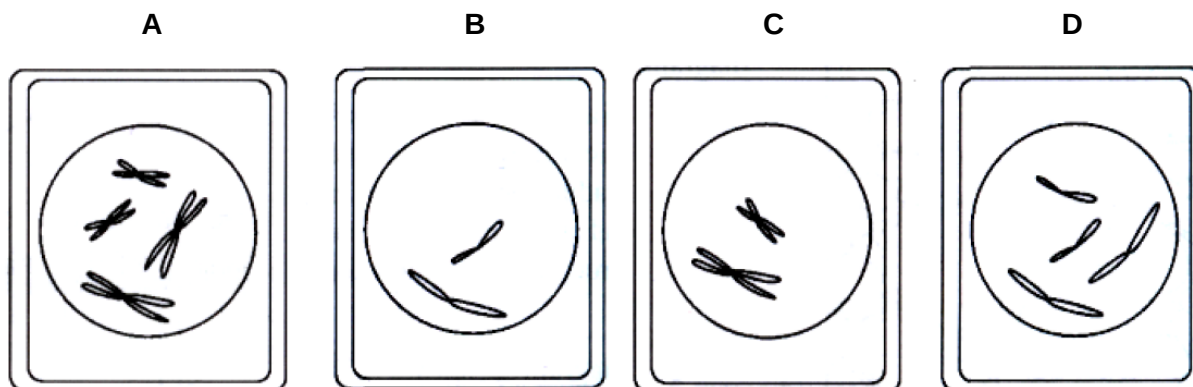


Which statement could **not** account for the differences in the levels of expression of globin chains?

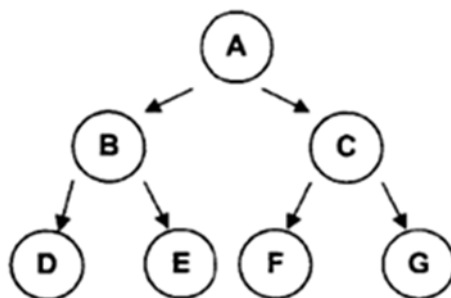
- A** A growth factor triggers the expression of a transcription factor that increases the rate of  $\beta$ -globin gene expression during the postnatal period.
- B** Alternative splicing results in the differences in the levels of expression of globin chains during the prenatal period.
- C** Methyl groups are added to regulatory sequences of  $\gamma$ -globin genes during the postnatal period, allowing for some proteins to bind.
- D** The shortening of poly(A) tail in the mRNA of  $\gamma$ -globin genes reduces its stability, resulting in a decrease in the rate of expression of  $\gamma$ -globin chains during the postnatal period.

- 12 A diploid cell contains four chromosomes.

Which diagram shows the nucleus at prophase of mitosis after a meiotic cell cycle?



- 13 The diagram shows cell A undergoing meiosis to produce four daughter cells, D, E, F and G.



If no crossing over occurred during meiosis, which cells would be genetically identical?

- A B and C
  - B D, E, F and G
  - C D and E; F and G
  - D none
- 14 Which event does **not** increase the chance of cancerous growth?
- A amplification of *p53* gene
  - B amplification of *ras* gene
  - C increase in telomerase activity
  - D loss of immunity

15 Which statements are **true** about all stem cells?

- 1 Stem cells can be induced to differentiate by environmental signals.
- 2 Stem cells are easily isolated and propagated.
- 3 Stem cells are able to develop into whole organisms if implanted into the womb.
- 4 Stem cells make more stem cells under appropriate conditions.

- A 1 and 4
- B 2 and 3
- C 1, 3 and 4
- D 1, 2, 3 and 4

16 Which observation best describes the process of natural selection?

- A change from simple to more complex organisms over time
- B change in size of the population over time
- C different rates of reproductive success of different genotypes
- D spontaneous occurrence of advantageous mutations

17 Which statement is **not** an example of a macroevolutionary process?

- A As a result of human activities, one bird species becomes extinct.
- B Birds and insects have wings that evolved separately.
- C One lion species splits to form two lion species over time.
- D Over a short period, the frequency of a single gene declines from 10% to 8%.



**18** A comparison of the following was made between human, rabbit, mouse, and chimpanzee:

- DNA coding sequence of the  $\beta$  globin gene
- DNA sequence in the introns of the  $\beta$  globin gene
- amino acid sequence of the  $\beta$  globin polypeptide.

The table shows the sequence similarity for the organisms being compared.

| organisms being compared | sequence similarity (%) |        |                     |
|--------------------------|-------------------------|--------|---------------------|
|                          | coding DNA              | intron | amino acid sequence |
| human and chimpanzee     | 100                     | 98.4   | 100                 |
| human and rabbit         | 89.3                    | 67     | 92.4                |
| human and mouse          | 82.1                    | 61     | 80.1                |

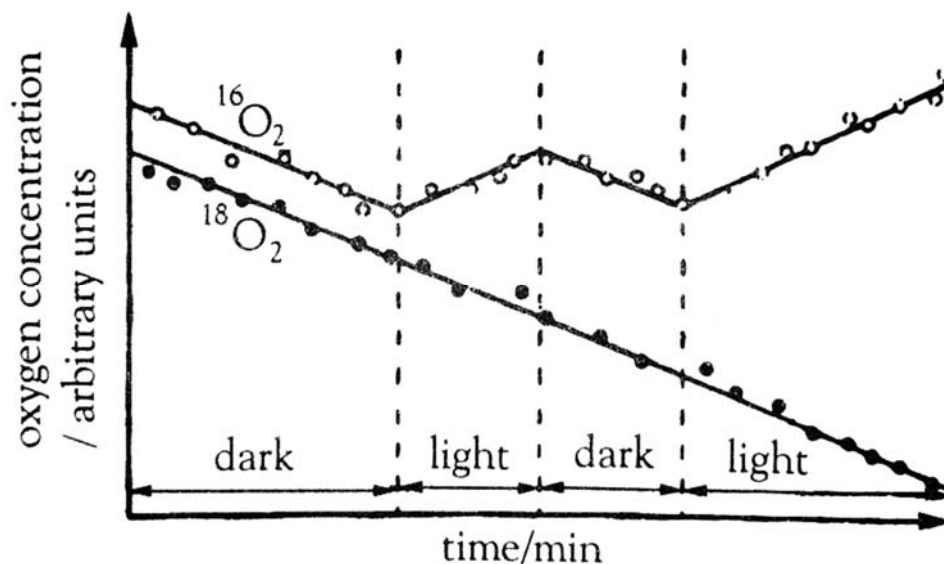
What could be concluded from the information given?

- A** A human is more closely related to a mouse than a rabbit.
- B** The comparison between human and rabbit indicates that the differences in their DNA did not always make a difference to the amino acid sequence.
- C** The variation between human and chimpanzee occurs in a region of the  $\beta$  globin gene that codes for amino acids.
- D** The variation in the intron sequence between human and mouse would account for some of the differences in the amino acid sequence.
- 19** Which region of the chloroplast has the lowest pH when sunlight shines on the organelle?
- A** intermembrane space
- B** starch grain
- C** stroma
- D** thylakoid space

- 20 Isotopes of oxygen can be used to distinguish between oxygen absorbed by plants and oxygen evolved.

A mixture of oxygen isotopes,  $^{16}\text{O}_2$  and  $^{18}\text{O}_2$ , was supplied to a suspension of the unicellular alga *Chlorella* which had previously been exposed to  $^{16}\text{O}_2$  only. During the following hour, changes in the concentration of these gases in the suspension were measured in light and dark conditions.

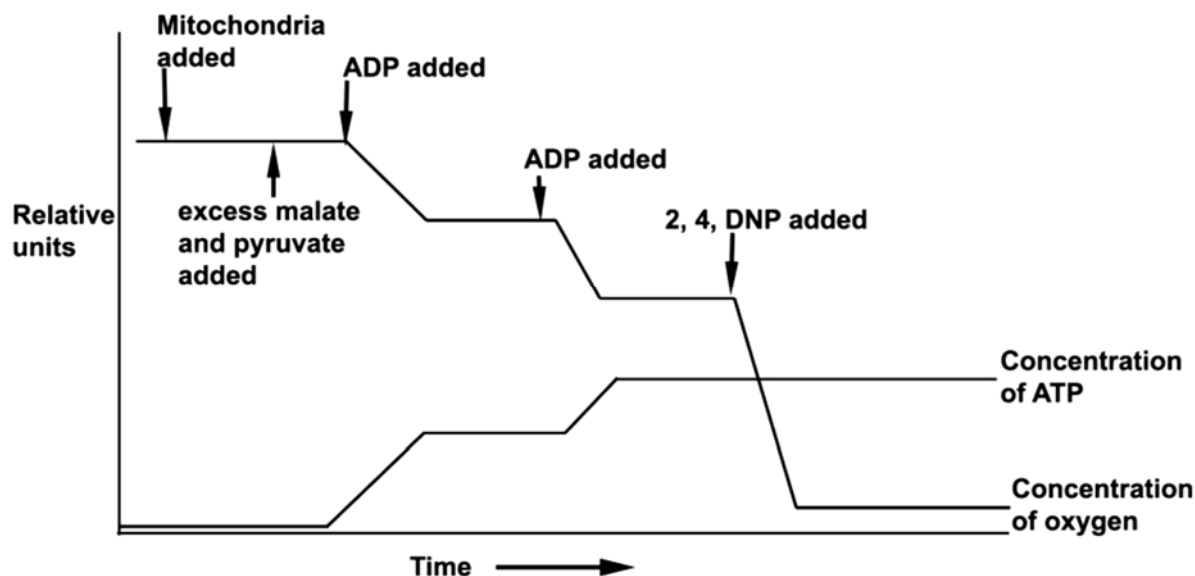
The graph shows the results.



What caused the concentration of  $^{16}\text{O}_2$  to rise in when light was provided?

- A  $^{18}\text{O}_2$  formed a decreasing proportion of the oxygen evolved.
- B  $^{16}\text{O}_2$  was absorbed at different rates in light and dark.
- C  $^{16}\text{O}_2$  was being produced in photosynthesis but was not being absorbed in respiration.
- D  $^{16}\text{O}_2$  was being produced in photosynthesis faster than it was being absorbed in respiration.
- 21 Which statement correctly describes a difference between aerobic and anaerobic respiration?
- A Lactate is formed in aerobic respiration in animals, whereas alcohol is formed in anaerobic respiration in plants.
- B Oxidation is complete in aerobic respiration, whereas oxidation is incomplete in anaerobic respiration.
- C Pyruvate is formed in aerobic respiration, whereas pyruvate is not formed in anaerobic respiration.
- D The energy released in aerobic respiration is stored in ATP, whereas the energy released in anaerobic respiration is not stored in ATP.

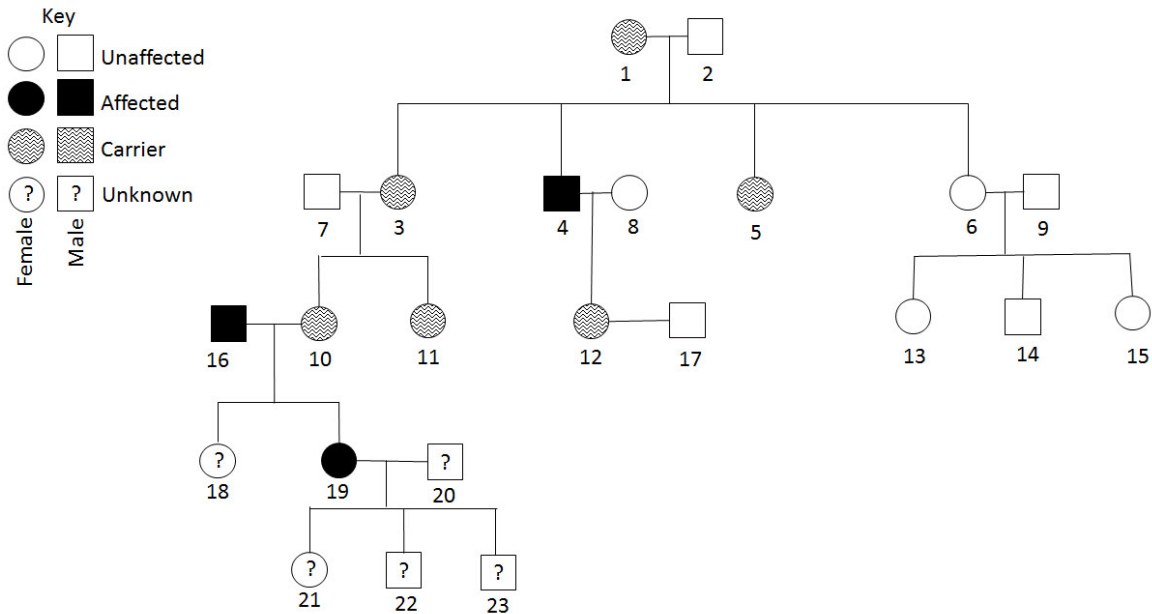
- 22 The graph shows the use of oxygen and the synthesis of ATP by isolated mitochondria upon the addition of various compounds.



What could be deduced from the graph?

- 1 ADP is not normally required for the process of respiration to occur.
  - 2 2, 4, DNP allows the process of respiration to proceed without ATP synthesis.
  - 3 The rate of respiration in this experiment is limited by the availability of malate and pyruvate.
- A 2 only  
 B 3 only  
 C 1 and 2 only  
 D 1 and 3 only
- 23 What is an example of a test cross?
- A dominant parent x hybrid  
 B dominant parent x recessive parent  
 C hybrid x hybrid  
 D hybrid x recessive parent

- 24 The pedigree shows the inheritance of an X-linked recessive trait.



What is the probability of individual 22 being affected?

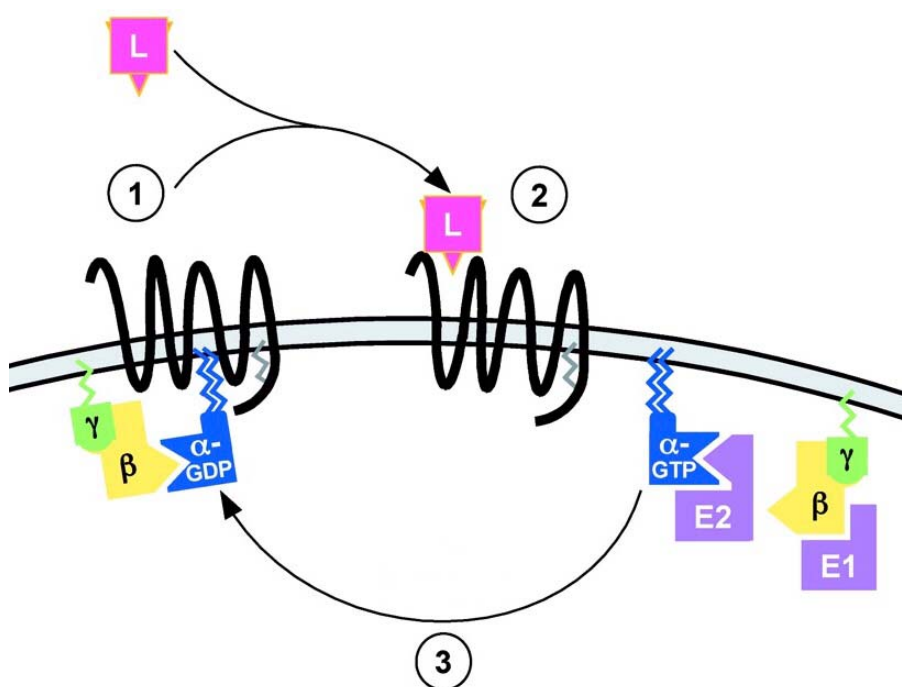
- A 0%  
 B 25%  
 C 50%  
 D 100%
- 25 Coat colour in some mammals is controlled by a gene with four alleles. The order of dominance for these alleles, in descending order, is as follows:

agouti (**C**), chinchilla (**C<sup>c</sup>**), Himalayan (**C<sup>h</sup>**), albino (**c**).

How many different genotypes for coat colour are possible?

- A 8                      B 10                      C 12                      D 16
- 26 Which enzyme is **not** involved in the regulation of signal transduction pathways?

- A GTPase  
 B kinase  
 C phosphatase  
 D phosphorylase



- 1 A heterotrimeric G-protein is associated with an active cell surface receptor that has seven transmembrane domains.
- 2 The binding of ligand L to the G-protein linked receptor is required for the activation of downstream effectors, E1 and E2.
- 3 A molecule of GDP displaces the GTP on the alpha subunit of the G-protein, causing the alpha subunit to reassociate with the beta-gamma complex.

- A** 1 only  
**B** 2 only  
**C** 1 and 3 only  
**D** 1, 2 and 3

**A** Antibodies are built from equal numbers of large (heavy) and small (light) peptide chains.

**B** Antibodies are secreted and function away from the cell.

**C** Antibodies have multiple identical antigen binding sites.

**D** Antibodies have heavy chains that determine their isotypes.

- 29 What describes a virus such as influenza which emerges suddenly and spreads globally?
- A endemic
  - B epidemic
  - C pandemic
  - D zoonotic
- 30 Which data is **not** an evidence of climate change?
- A carbon dioxide concentration measured from air bubbles trapped in an ice core from the Antarctic
  - B changes in glacier formation and melting through photographs and maps
  - C maximum temperatures recorded during summer each year
  - D analysis of pollens from plants preserved in different layers of lake bed sediments